# Expected Outputs and Evaluation Rubrics for the Hackathon Problem 1 (Siemens - GeoCAiM)

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## 1 Introduction

This document outlines the expected outputs and evaluation rubrics for the Hackathon Problem 1 (Siemens - GeoCAiM).

## 2 Expected Outputs: Round 1

The expected outputs include a technical submission along with a short 3-min presentation video for Round 1. The technical solution includes three sets of outputs classified as easy, medium, and hard, along with a brief description of the methods, approaches, and techniques used to solve the problem including any code developed. The outputs should be submitted in the specified format and following conventions.

The outputs are classified as follows:

- Easy: Ordered sequence of main and sub-operations for planar CNC machining (.json)
- Medium: IPW after each operation (.stl)
- Hard: Tool type, Tool size, and Tool path for each operation along with machining parameters represented as NC code/G-code for each operation (.ptp)

A helper script is provided to ensure that the outputs are in the correct format. The script will check for the following:

- Correct file format and proper naming conventions for files (e.g., .json, .stl, .ptp, etc.)
- Formatting of the output files
- Presence of all required fields in the output
- Validity of data types and ranges

# 3 Overall Scoring and Evaluation Rubrics

The evaluation rubrics are designed to assess the outputs and score. Adherence to the specified format and conventions is needed for the outputs to be considered valid. Incorrectly formatted solutions may be disqualified from scoring. The judges can penalize the participants for engaging in unethical practices against the spirit of the competition at their discretion. The scoring will be based on the following criteria:

## 3.1 Evaluation of Outputs - Round 1

### 3.1.1 Easy <20 points>

Sequence accuracy will be evaluated using the normalized Levenshtein distance and the F1 score, with 10 points each for a total of 20 points.

The normalized Levenshtein distance is computed as $d_L(P, G)/\max(|P|, |G|)$, where $d_L$ is the Levenshtein edit distance between the predicted operation sequence $P$ and the ground truth sequence $G$, and $|\cdot|$ denotes the number of operations in the sequence. Lower values indicate better reproduction of the *order* of the operations. The F1 score is the harmonic mean of precision and recall computed over the multiset of predicted operations against the ground truth operations. The two sub-scores are added to give the score for the easy output. The scoring for each will be as follows:

| Normalized Levenshtein Distance (lower is better) | F1 Score | Points (each) |
|---|---|---|
| 0.0 - 0.1 | 0.95 - 1.0 | 10 |
| 0.1 - 0.2 | 0.85 - 0.95 | 8 |
| 0.2 - 0.3 | 0.75 - 0.85 | 6 |
| 0.3 - 0.4 | 0.65 - 0.75 | 4 |
| 0.4 - 0.5 | 0.55 - 0.65 | 2 |
| 0.5 - 1.0 | 0.0 - 0.55 | 0 |

### 3.1.2 Medium <35 points>

The IPWs will be evaluated using Intersection over Union (IoU) metric. The IoU will be calculated between the predicted IPW and the ground truth IPW for each operation. In addition final IPW will be compared against the target geometry (BRep). The IoU score will be averaged across all operations to obtain a final score for the medium output. The scoring will be as follows:

### 3.1.3 Hard <45 points>

The hard outputs will be evaluated based on the correctness and completeness of the tool type, tool sizes and tool path selection, as well as the accuracy of the tool path represented

| IoU Score | Points |
|---|---|
| 0.999 - 1.0 | 35 |
| 0.99 - 0.999 | 25 |
| 0.98 - 0.99 | 20 |
| 0.95 - 0.98 | 15 |
| 0.90 - 0.95 | 10 |
| 0.0 - 0.90 | 0 |

as NC code/G-code. The submitted G-code is *not* compared textually against the ground truth, since many syntactically different programs describe the same machining operation.

**Tool selection (type and diameter) (20 points).** For each operation the submitted tool is compared against the ground truth tool on two attributes. The tool *type* (e.g. end mill, chamfer mill, drill, etc.) must match exactly. The tool *size* is compared on its principal dimensions, cutting diameter $D$, and is accepted if diameter lies within a relative tolerance of the ground truth value. A wrong tool type scores zero for the whole operation, as the resulting geometry cannot be equivalent.

| Tool type match | Max. relative size error $\lvert d'_i - d_i \rvert / d_i$ | Points |
|---|---|---|
| Exact | 0.00 - 0.02 | 10 |
| Exact | 0.02 - 0.05 | 8 |
| Exact | 0.05 - 0.10 | 6 |
| Exact | 0.10 - 0.20 | 3 |
| Exact | $> 0.20$ | 1 |
| Mismatch | any | 0 |

**Tool path geometry (25 points).** The tool path is evaluated by comparing a predicted CNC tool path's swept volume against the boolean difference of before/after IPW meshes. The volume comparison is made through IOU of the swept volume and the ground truth. We also compute the over/under cut percentage of the volume by the tool path. Overcut refers to the volume of material removed that should not have been removed, while undercut refers to the volume of material that should have been removed but was not. The scoring will be as follows:

| IOU | Points | Over/Under cut % | Points (each) |
|---|---|---|---|
| 0.99 - 1.0 | 15 | 0.0 - 0.05 | 7.5 |
| 0.95 - 0.99 | 12 | 0.05 - 0.15 | 6 |
| 0.85 - 0.95 | 9 | 0.15 - 0.25 | 4.5 |
| 0.75 - 0.85 | 6 | 0.25 - 0.35 | 3 |
| 0.65 - 0.75 | 3 | 0.35 - 0.45 | 1.5 |
| 0.0 - 0.65 | 0 | 0.45 - 1.0 | 0 |

### 3.2 Evaluation of Methods and Presentation - Round 2

In round two, the evaluation will also focus on the methods, approaches, and techniques used to solve the problem along with the technical score from round one. The methods used in the submission will be evaluated and judged on the following criteria:

- **Originality and creativity of the approach**: Provide a brief literature review of the methods, including any relevant references. Explain whether your approach differs from existing methods and any adaptations or improvements made.
- **Effectiveness and computational efficiency of the methods used**: Describe the performance of your methods in terms of speed, resource usage, and scalability. Compare with any baselines or existing methods if applicable.
- **Clarity and completeness of the documentation provided**: Include any training code, scripts, or notebooks used to generate the outputs. The documentation should be clear enough for someone else to reproduce the results. Explain the rationale behind the chosen methods, any assumptions made.
- **Generalizability of the results**: Discuss the applicability of your methods to different datasets, scenarios, or problem settings. Highlight any limitations or constraints. Document any cases the methods may not work well and suggest potential improvements or future work.
- **Scientific integrity and rigor**: Reverse engineering or hacky solutions will be penalized.

## 4 Note on Use of LLMs/VLMs and Pre-trained Models

You are free to use any models for writing code or documentation. If any of the outputs (easy, medium, or hard) are generated directly using LLMs/VLMs or pre-trained vision models, the following information must be provided:

- The name and version of the model used
- The source or repository from which the model was obtained
- Any modifications made to the model or fine tuning of its parameters
- A brief description of how the model was used in generating the outputs